

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15E

SPECTROSCOPY & NMR (CHROMATOGRAPHY, MASS SPEC, ^{13}C NMR, ^1H NMR & SPLITTING)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E: Spectroscopy & NMR (Chromatography, Mass Spec, ^{13}C NMR, ^1H NMR & Splitting)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	229 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. ¹H NMR Spectrum Analysis of Ethyl Ethanoate****[5 Marks]**
WCH14/01/Jan23/Q23

Ethyl ethanoate, CH₃COOCH₂CH₃, is analysed using high-resolution ¹H NMR spectroscopy.

- (a) Predict the number of peaks (environments) in the ¹H NMR spectrum of ethyl ethanoate. [1]
- (b) Describe the splitting pattern (n+1 rule) and relative area for each peak. [4]

Q2. 2. Mass Spectrometry Fragmentation & D₂O Shake**[5 Marks]**
WCH14/01/Oct22/Q24

Compound X (C₃H₆O₂) has an IR absorption at 1715 cm⁻¹ and a broad absorption at 2500–3300 cm⁻¹.

- (a) Identify the functional group present in X from the IR data. [1]
- (b) State the effect of adding D₂O (D₂O shake) on the ¹H NMR spectrum of X. [2]
- (c) Identify the fragment ion responsible for the peak at m/z = 45 in the mass spectrum of X. [2]

Q3. 3. ¹³C NMR & Chromatography Step 3**[3 Marks]**
WCH14/01/Sample/Q3

Consider organic compound 3 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
- (b) State how GC-MS separates and identifies components in a mixture. [1]

Q4. 4. ¹³C NMR & Chromatography Step 4**[3 Marks]**
WCH14/01/Sample/Q4

Consider organic compound 4 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
- (b) State how GC-MS separates and identifies components in a mixture. [1]

Q5. 5. ¹³C NMR & Chromatography Step 5**[3 Marks]**
WCH14/01/Sample/Q5

Consider organic compound 5 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
- (b) State how GC-MS separates and identifies components in a mixture. [1]

Q6. 6. ¹³C NMR & Chromatography Step 6**[3 Marks]**
WCH14/01/Sample/Q6

Consider organic compound 6 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
- (b) State how GC-MS separates and identifies components in a mixture. [1]

Q7. 7. ¹³C NMR & Chromatography Step 7**[3 Marks]**
WCH14/01/Sample/Q7

Consider organic compound 7 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
- (b) State how GC-MS separates and identifies components in a mixture. [1]

Q8. 8. ¹³C NMR & Chromatography Step 8**[3 Marks]**
WCH14/01/Sample/Q8

Consider organic compound 8 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q9. 9. ^{13}C NMR & Chromatography Step 9

[3 Marks]
WCH14/01/Sample/Q9

Consider organic compound 9 analysed by ^{13}C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q10. 10. ^{13}C NMR & Chromatography Step 10

[3 Marks]
WCH14/01/Sample/Q10

Consider organic compound 10 analysed by ^{13}C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q11. 11. ^{13}C NMR & Chromatography Step 11

[3 Marks]
WCH14/01/Sample/Q11

Consider organic compound 11 analysed by ^{13}C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q12. 12. ^{13}C NMR & Chromatography Step 12

[3 Marks]
WCH14/01/Sample/Q12

Consider organic compound 12 analysed by ^{13}C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q13. 13. ^{13}C NMR & Chromatography Step 13

[3 Marks]
WCH14/01/Sample/Q13

Consider organic compound 13 analysed by ^{13}C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q14. 14. ^{13}C NMR & Chromatography Step 14

[3 Marks]
WCH14/01/Sample/Q14

Consider organic compound 14 analysed by ^{13}C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q15. 15. ^{13}C NMR & Chromatography Step 15

[3 Marks]
WCH14/01/Sample/Q15

Consider organic compound 15 analysed by ^{13}C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q16. 16. ^{13}C NMR & Chromatography Step 16

[3 Marks]
WCH14/01/Sample/Q16

Consider organic compound 16 analysed by ^{13}C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]

- (b) State how GC-MS separates and identifies components in a mixture. [1]

Q17. 17. ¹³C NMR & Chromatography Step 17**[3 Marks]****WCH14/01/Sample/Q17**

Consider organic compound 17 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q18. 18. ¹³C NMR & Chromatography Step 18**[3 Marks]****WCH14/01/Sample/Q18**

Consider organic compound 18 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q19. 19. ¹³C NMR & Chromatography Step 19**[3 Marks]****WCH14/01/Sample/Q19**

Consider organic compound 19 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q20. 20. ¹³C NMR & Chromatography Step 20**[3 Marks]****WCH14/01/Sample/Q20**

Consider organic compound 20 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q21. 21. ¹³C NMR & Chromatography Step 21**[3 Marks]****WCH14/01/Sample/Q21**

Consider organic compound 21 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q22. 22. ¹³C NMR & Chromatography Step 22**[3 Marks]****WCH14/01/Sample/Q22**

Consider organic compound 22 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q23. 23. ¹³C NMR & Chromatography Step 23**[3 Marks]****WCH14/01/Sample/Q23**

Consider organic compound 23 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q24. 24. ¹³C NMR & Chromatography Step 24**[3 Marks]****WCH14/01/Sample/Q24**

Consider organic compound 24 analysed by ¹³C NMR and GC-MS.

- (a) Deduce the number of peaks in its ¹³C NMR spectrum based on symmetry. [2]
(b) State how GC-MS separates and identifies components in a mixture. [1]

Q25. 25. ^{13}C NMR & Chromatography Step 25**[3 Marks]****WCH14/01/Sample/Q25**

Consider organic compound 25 analysed by ^{13}C NMR and GC-MS.

- (a) Deduce the number of peaks in its ^{13}C NMR spectrum based on symmetry. [2]
- (b) State how GC-MS separates and identifies components in a mixture. [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Multi-Spectral Structural Elucidation**
26**[6 Marks]**
WCH14/01/Hard/Q26

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q27. 27. A* Challenge: Multi-Spectral Structural Elucidation 27**[6 Marks]**
WCH14/01/Hard/Q27

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q28. 28. A* Challenge: Multi-Spectral Structural Elucidation
28**[6 Marks]**
WCH14/01/Hard/Q28

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q29. 29. A* Challenge: Multi-Spectral Structural Elucidation
29**[6 Marks]**
WCH14/01/Hard/Q29

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q30. 30. A* Challenge: Multi-Spectral Structural Elucidation
30**[6 Marks]**
WCH14/01/Hard/Q30

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q31. 31. A* Challenge: Multi-Spectral Structural Elucidation 31**[6 Marks]**
WCH14/01/Hard/Q31

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q32. 32. A* Challenge: Multi-Spectral Structural Elucidation**32****[6 Marks]****WCH14/01/Hard/Q32**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q33. 33. A* Challenge: Multi-Spectral Structural Elucidation**33****[6 Marks]****WCH14/01/Hard/Q33**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q34. 34. A* Challenge: Multi-Spectral Structural Elucidation**34****[6 Marks]****WCH14/01/Hard/Q34**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q35. 35. A* Challenge: Multi-Spectral Structural Elucidation**35****[6 Marks]****WCH14/01/Hard/Q35**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q36. 36. A* Challenge: Multi-Spectral Structural Elucidation**36****[6 Marks]****WCH14/01/Hard/Q36**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

Q37. 37. A* Challenge: Multi-Spectral Structural Elucidation 37

[6 Marks]
WCH14/01/Hard/Q37

Compound Y (C₄H₈O₂) has the following spectroscopic data:

- IR: Strong C=O absorption at 1740 cm⁻¹ (ester).
- ¹³C NMR: 4 distinct peaks at δ = 14, 21, 60, 171 ppm.
- ¹H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

(a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

Q38. 38. A* Challenge: Multi-Spectral Structural Elucidation 38

[6 Marks]
WCH14/01/Hard/Q38

Compound Y (C₄H₈O₂) has the following spectroscopic data:

- IR: Strong C=O absorption at 1740 cm⁻¹ (ester).
- ¹³C NMR: 4 distinct peaks at δ = 14, 21, 60, 171 ppm.
- ¹H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

(a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

Q39. 39. A* Challenge: Multi-Spectral Structural Elucidation 39

[6 Marks]
WCH14/01/Hard/Q39

Compound Y (C₄H₈O₂) has the following spectroscopic data:

- IR: Strong C=O absorption at 1740 cm⁻¹ (ester).
- ¹³C NMR: 4 distinct peaks at δ = 14, 21, 60, 171 ppm.
- ¹H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

(a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

Q40. 40. A* Challenge: Multi-Spectral Structural Elucidation 40

[6 Marks]
WCH14/01/Hard/Q40

Compound Y (C₄H₈O₂) has the following spectroscopic data:

- IR: Strong C=O absorption at 1740 cm⁻¹ (ester).
- ¹³C NMR: 4 distinct peaks at δ = 14, 21, 60, 171 ppm.
- ¹H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

(a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

Q41. 41. A* Challenge: Multi-Spectral Structural Elucidation 41

[6 Marks]
WCH14/01/Hard/Q41

Compound Y (C₄H₈O₂) has the following spectroscopic data:

- IR: Strong C=O absorption at 1740 cm⁻¹ (ester).
- ¹³C NMR: 4 distinct peaks at δ = 14, 21, 60, 171 ppm.
- ¹H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

(a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

Q42. 42. A* Challenge: Multi-Spectral Structural Elucidation 42

[6 Marks]
WCH14/01/Hard/Q42

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q43. 43. A* Challenge: Multi-Spectral Structural Elucidation**43****[6 Marks]****WCH14/01/Hard/Q43**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q44. 44. A* Challenge: Multi-Spectral Structural Elucidation**44****[6 Marks]****WCH14/01/Hard/Q44**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q45. 45. A* Challenge: Multi-Spectral Structural Elucidation**45****[6 Marks]****WCH14/01/Hard/Q45**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q46. 46. A* Challenge: Multi-Spectral Structural Elucidation**46****[6 Marks]****WCH14/01/Hard/Q46**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]
(b) Give the systematic IUPAC name for compound Y. [1]

Q47. 47. A* Challenge: Multi-Spectral Structural Elucidation**47****[6 Marks]****WCH14/01/Hard/Q47**

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

- (a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

Q48. 48. A* Challenge: Multi-Spectral Structural Elucidation
48

[6 Marks]
WCH14/01/Hard/Q48

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

(a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

Q49. 49. A* Challenge: Multi-Spectral Structural Elucidation
49

[6 Marks]
WCH14/01/Hard/Q49

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

(a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

Q50. 50. A* Challenge: Multi-Spectral Structural Elucidation
50

[6 Marks]
WCH14/01/Hard/Q50

Compound Y ($C_4H_8O_2$) has the following spectroscopic data:

- IR: Strong $C=O$ absorption at 1740 cm^{-1} (ester).
- ^{13}C NMR: 4 distinct peaks at $\delta = 14, 21, 60, 171\text{ ppm}$.
- 1H NMR: Singlet at 2.0 ppm (3H), Quartet at 4.1 ppm (2H), Triplet at 1.2 ppm (3H).

(a) Analyze each piece of spectroscopic data to deduce the complete structural formula of compound Y. [5]

(b) Give the systematic IUPAC name for compound Y. [1]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15E

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. ¹ H NMR Spectrum Analysis of Ethyl Ethanoate 1. (a) 3 peaks (3 proton environments) (1). 1. (b) CH ₃ CO- protons: singlet (0 adjacent H, n=0 → 1 peak), integration = 3H (1). -OCH ₂ - protons: quartet (3 adjacent H on CH ₃ , n=3 → 4 peaks), integration = 2H (1). -CH ₃ protons of ethyl: triplet (2 adjacent H on CH ₂ , n=2 → 3 peaks), integration = 3H (1).	[5]
Q2	Q2: 2. Mass Spectrometry Fragmentation & D ₂ O Shake 2. (a) Carboxylic acid (-COOH) group (C=O at 1715, O-H at 2500-3300) (1). 2. (b) The acidic -OH proton exchanges with deuterium (-OH + D ₂ O → -OD + HOD); the -OH peak at delta ~11 ppm disappears (2). 2. (c) [COOH] ⁺ ion (m/z = 12 + 32 + 1 = 45) (2).	[5]
Q3	Q3: 3. ¹³ C NMR & Chromatography Step 3 3. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 3. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q4	Q4: 4. ¹³ C NMR & Chromatography Step 4 4. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 4. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q5	Q5: 5. ¹³ C NMR & Chromatography Step 5 5. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 5. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q6	Q6: 6. ¹³ C NMR & Chromatography Step 6 6. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 6. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q7	Q7: 7. ¹³ C NMR & Chromatography Step 7 7. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 7. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q8	Q8: 8. ¹³ C NMR & Chromatography Step 8 8. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 8. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q9	Q9: 9. ¹³ C NMR & Chromatography Step 9 9. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 9. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q10	Q10: 10. ¹³ C NMR & Chromatography Step 10 10. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 10. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q11	Q11: 11. ¹³ C NMR & Chromatography Step 11 11. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 11. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q12	Q12: 12. ¹³ C NMR & Chromatography Step 12 12. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 12. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]

Q13	Q13: 13. 13C NMR & Chromatography Step 13 13. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 13. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q14	Q14: 14. 13C NMR & Chromatography Step 14 14. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 14. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q15	Q15: 15. 13C NMR & Chromatography Step 15 15. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 15. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q16	Q16: 16. 13C NMR & Chromatography Step 16 16. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 16. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q17	Q17: 17. 13C NMR & Chromatography Step 17 17. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 17. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q18	Q18: 18. 13C NMR & Chromatography Step 18 18. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 18. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q19	Q19: 19. 13C NMR & Chromatography Step 19 19. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 19. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q20	Q20: 20. 13C NMR & Chromatography Step 20 20. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 20. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q21	Q21: 21. 13C NMR & Chromatography Step 21 21. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 21. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q22	Q22: 22. 13C NMR & Chromatography Step 22 22. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 22. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q23	Q23: 23. 13C NMR & Chromatography Step 23 23. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 23. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q24	Q24: 24. 13C NMR & Chromatography Step 24 24. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 24. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]
Q25	Q25: 25. 13C NMR & Chromatography Step 25 25. (a) Count non-equivalent carbon environments considering molecular symmetry (2). 25. (b) GC separates components by retention time; MS identifies each component by its molecular ion and fragmentation pattern (1).	[3]

Q26	Q26: 26. A* Challenge: Multi-Spectral Structural Elucidation 26 26. (a) IR 1740 cm⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH₃CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH₂CH₃ ethyl group (1). Combined structure = CH₃COOCH₂CH₃ (1). 26. (b) Ethyl ethanoate (1).	[6]
Q27	Q27: 27. A* Challenge: Multi-Spectral Structural Elucidation 27 27. (a) IR 1740 cm⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH₃CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH₂CH₃ ethyl group (1). Combined structure = CH₃COOCH₂CH₃ (1). 27. (b) Ethyl ethanoate (1).	[6]
Q28	Q28: 28. A* Challenge: Multi-Spectral Structural Elucidation 28 28. (a) IR 1740 cm⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH₃CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH₂CH₃ ethyl group (1). Combined structure = CH₃COOCH₂CH₃ (1). 28. (b) Ethyl ethanoate (1).	[6]
Q29	Q29: 29. A* Challenge: Multi-Spectral Structural Elucidation 29 29. (a) IR 1740 cm⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH₃CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH₂CH₃ ethyl group (1). Combined structure = CH₃COOCH₂CH₃ (1). 29. (b) Ethyl ethanoate (1).	[6]
Q30	Q30: 30. A* Challenge: Multi-Spectral Structural Elucidation 30 30. (a) IR 1740 cm⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH₃CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH₂CH₃ ethyl group (1). Combined structure = CH₃COOCH₂CH₃ (1). 30. (b) Ethyl ethanoate (1).	[6]
Q31	Q31: 31. A* Challenge: Multi-Spectral Structural Elucidation 31 31. (a) IR 1740 cm⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH₃CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH₂CH₃ ethyl group (1). Combined structure = CH₃COOCH₂CH₃ (1). 31. (b) Ethyl ethanoate (1).	[6]
Q32	Q32: 32. A* Challenge: Multi-Spectral Structural Elucidation 32 32. (a) IR 1740 cm⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH₃CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH₂CH₃ ethyl group (1). Combined structure = CH₃COOCH₂CH₃ (1). 32. (b) Ethyl ethanoate (1).	[6]
Q33	Q33: 33. A* Challenge: Multi-Spectral Structural Elucidation 33 33. (a) IR 1740 cm⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH₃CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH₂CH₃ ethyl group (1). Combined structure = CH₃COOCH₂CH₃ (1). 33. (b) Ethyl ethanoate (1).	[6]
Q34	Q34: 34. A* Challenge: Multi-Spectral Structural Elucidation 34 34. (a) IR 1740 cm⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH₃CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH₂CH₃ ethyl group (1). Combined structure = CH₃COOCH₂CH₃ (1). 34. (b) Ethyl ethanoate (1).	[6]

Q35	Q35: 35. A* Challenge: Multi-Spectral Structural Elucidation 35 35. (a) IR 1740 cm ⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH ₃ CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH ₂ CH ₃ ethyl group (1). Combined structure = CH ₃ COOCH ₂ CH ₃ (1). 35. (b) Ethyl ethanoate (1).	[6]
Q36	Q36: 36. A* Challenge: Multi-Spectral Structural Elucidation 36 36. (a) IR 1740 cm ⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH ₃ CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH ₂ CH ₃ ethyl group (1). Combined structure = CH ₃ COOCH ₂ CH ₃ (1). 36. (b) Ethyl ethanoate (1).	[6]
Q37	Q37: 37. A* Challenge: Multi-Spectral Structural Elucidation 37 37. (a) IR 1740 cm ⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH ₃ CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH ₂ CH ₃ ethyl group (1). Combined structure = CH ₃ COOCH ₂ CH ₃ (1). 37. (b) Ethyl ethanoate (1).	[6]
Q38	Q38: 38. A* Challenge: Multi-Spectral Structural Elucidation 38 38. (a) IR 1740 cm ⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH ₃ CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH ₂ CH ₃ ethyl group (1). Combined structure = CH ₃ COOCH ₂ CH ₃ (1). 38. (b) Ethyl ethanoate (1).	[6]
Q39	Q39: 39. A* Challenge: Multi-Spectral Structural Elucidation 39 39. (a) IR 1740 cm ⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH ₃ CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH ₂ CH ₃ ethyl group (1). Combined structure = CH ₃ COOCH ₂ CH ₃ (1). 39. (b) Ethyl ethanoate (1).	[6]
Q40	Q40: 40. A* Challenge: Multi-Spectral Structural Elucidation 40 40. (a) IR 1740 cm ⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH ₃ CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH ₂ CH ₃ ethyl group (1). Combined structure = CH ₃ COOCH ₂ CH ₃ (1). 40. (b) Ethyl ethanoate (1).	[6]
Q41	Q41: 41. A* Challenge: Multi-Spectral Structural Elucidation 41 41. (a) IR 1740 cm ⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH ₃ CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH ₂ CH ₃ ethyl group (1). Combined structure = CH ₃ COOCH ₂ CH ₃ (1). 41. (b) Ethyl ethanoate (1).	[6]
Q42	Q42: 42. A* Challenge: Multi-Spectral Structural Elucidation 42 42. (a) IR 1740 cm ⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH ₃ CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH ₂ CH ₃ ethyl group (1). Combined structure = CH ₃ COOCH ₂ CH ₃ (1). 42. (b) Ethyl ethanoate (1).	[6]
Q43	Q43: 43. A* Challenge: Multi-Spectral Structural Elucidation 43 43. (a) IR 1740 cm ⁻¹ confirms ester C=O (1). 13C NMR 4 peaks confirms 4 distinct carbon environments (1). 1H NMR singlet at 2.0 ppm (3H) = CH ₃ CO- group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = -OCH ₂ CH ₃ ethyl group (1). Combined structure = CH ₃ COOCH ₂ CH ₃ (1). 43. (b) Ethyl ethanoate (1).	[6]

Q44	Q44: 44. A* Challenge: Multi-Spectral Structural Elucidation 44 44. (a) IR 1740 cm^{-1} confirms ester $\text{C}=\text{O}$ (1). ^{13}C NMR 4 peaks confirms 4 distinct carbon environments (1). ^1H NMR singlet at 2.0 ppm (3H) = $\text{CH}_3\text{CO}-$ group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = $-\text{OCH}_2\text{CH}_3$ ethyl group (1). Combined structure = $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (1). 44. (b) Ethyl ethanoate (1).	[6]
Q45	Q45: 45. A* Challenge: Multi-Spectral Structural Elucidation 45 45. (a) IR 1740 cm^{-1} confirms ester $\text{C}=\text{O}$ (1). ^{13}C NMR 4 peaks confirms 4 distinct carbon environments (1). ^1H NMR singlet at 2.0 ppm (3H) = $\text{CH}_3\text{CO}-$ group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = $-\text{OCH}_2\text{CH}_3$ ethyl group (1). Combined structure = $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (1). 45. (b) Ethyl ethanoate (1).	[6]
Q46	Q46: 46. A* Challenge: Multi-Spectral Structural Elucidation 46 46. (a) IR 1740 cm^{-1} confirms ester $\text{C}=\text{O}$ (1). ^{13}C NMR 4 peaks confirms 4 distinct carbon environments (1). ^1H NMR singlet at 2.0 ppm (3H) = $\text{CH}_3\text{CO}-$ group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = $-\text{OCH}_2\text{CH}_3$ ethyl group (1). Combined structure = $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (1). 46. (b) Ethyl ethanoate (1).	[6]
Q47	Q47: 47. A* Challenge: Multi-Spectral Structural Elucidation 47 47. (a) IR 1740 cm^{-1} confirms ester $\text{C}=\text{O}$ (1). ^{13}C NMR 4 peaks confirms 4 distinct carbon environments (1). ^1H NMR singlet at 2.0 ppm (3H) = $\text{CH}_3\text{CO}-$ group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = $-\text{OCH}_2\text{CH}_3$ ethyl group (1). Combined structure = $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (1). 47. (b) Ethyl ethanoate (1).	[6]
Q48	Q48: 48. A* Challenge: Multi-Spectral Structural Elucidation 48 48. (a) IR 1740 cm^{-1} confirms ester $\text{C}=\text{O}$ (1). ^{13}C NMR 4 peaks confirms 4 distinct carbon environments (1). ^1H NMR singlet at 2.0 ppm (3H) = $\text{CH}_3\text{CO}-$ group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = $-\text{OCH}_2\text{CH}_3$ ethyl group (1). Combined structure = $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (1). 48. (b) Ethyl ethanoate (1).	[6]
Q49	Q49: 49. A* Challenge: Multi-Spectral Structural Elucidation 49 49. (a) IR 1740 cm^{-1} confirms ester $\text{C}=\text{O}$ (1). ^{13}C NMR 4 peaks confirms 4 distinct carbon environments (1). ^1H NMR singlet at 2.0 ppm (3H) = $\text{CH}_3\text{CO}-$ group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = $-\text{OCH}_2\text{CH}_3$ ethyl group (1). Combined structure = $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (1). 49. (b) Ethyl ethanoate (1).	[6]
Q50	Q50: 50. A* Challenge: Multi-Spectral Structural Elucidation 50 50. (a) IR 1740 cm^{-1} confirms ester $\text{C}=\text{O}$ (1). ^{13}C NMR 4 peaks confirms 4 distinct carbon environments (1). ^1H NMR singlet at 2.0 ppm (3H) = $\text{CH}_3\text{CO}-$ group (1). Quartet at 4.1 ppm (2H) and triplet at 1.2 ppm (3H) = $-\text{OCH}_2\text{CH}_3$ ethyl group (1). Combined structure = $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (1). 50. (b) Ethyl ethanoate (1).	[6]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 15E**FAQ #1: n+1 Splitting Rule Application****Category: 1H NMR Splitting****COMMON EXAMINER TRAP / MISTAKE:**

Counting protons on the same carbon atom when determining splitting.

MODEL ANSWER / EXAMINER EXPECTATION:

The n+1 rule states that a proton peak is split into n+1 sub-peaks, where n is the number of protons on ADJACENT carbon atoms (3 bonds away). Protons on the same carbon do not split each other.

FAQ #2: D2O Exchange Mechanism**Category: D2O Shake****COMMON EXAMINER TRAP / MISTAKE:**

Expecting C-H protons to exchange with D2O.

MODEL ANSWER / EXAMINER EXPECTATION:

D2O shake ONLY removes peaks for labile -OH and -NH protons because they exchange rapidly with deuterium (-OH + D2O → -OD + HOD). Protons attached directly to carbon (C-H) do NOT exchange.

FAQ #3: TMS Reference Peak**Category: NMR Standard****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting the chemical shift value of TMS.

MODEL ANSWER / EXAMINER EXPECTATION:

Tetramethylsilane Si(CH₃)₄ (TMS) is assigned a chemical shift of EXACTLY 0.0 ppm. Its 12 identical H atoms give a sharp reference peak at the far right of the spectrum.

FAQ #4: 13C NMR Integration Traces**Category: 13C NMR vs 1H NMR****COMMON EXAMINER TRAP / MISTAKE:**

Assuming peak heights in 13C NMR represent the number of carbons.

MODEL ANSWER / EXAMINER EXPECTATION:

13C NMR peak heights/areas are NOT strictly proportional to the number of carbon atoms (unlike 1H NMR integration). ONLY the number of peaks (carbon environments) and chemical shift values are evaluated.

FAQ #5: Mass Spec Molecular Ion Peak Identification**Category: Mass Spectrometry****COMMON EXAMINER TRAP / MISTAKE:**

Confusing the molecular ion M⁺ peak with fragment peaks.

MODEL ANSWER / EXAMINER EXPECTATION:

The molecular ion M⁺ peak is the peak with the HIGHEST m/z value (excluding small M+1 isotope peaks). Its m/z value gives the relative molecular mass (Mr) of the compound.

FAQ #6: Mass Spec Fragment Ion Positive Charge Requirement**Category: Mass Spec Formulas****COMMON EXAMINER TRAP / MISTAKE:**

Writing fragment species without a positive charge.

MODEL ANSWER / EXAMINER EXPECTATION:

Mass spectrometers ONLY detect POSITIVELY CHARGED ions. Fragment formulas MUST include a positive charge (e.g. $[\text{CH}_3\text{CO}]^+$, $[\text{C}_2\text{H}_5]^+$, $[\text{COOH}]^+$).

FAQ #7: Infrared O–H Absorption Differences**Category: IR Spectroscopy****COMMON EXAMINER TRAP / MISTAKE:**

Confusing alcohol O–H with carboxylic acid O–H absorption.

MODEL ANSWER / EXAMINER EXPECTATION:

Alcohol O–H absorbs as a smooth broad peak at $3200\text{--}3600\text{ cm}^{-1}$. Carboxylic acid O–H absorbs as a VERY BROAD, distorted peak at $2500\text{--}3300\text{ cm}^{-1}$ overlapping the C–H region.

FAQ #8: Deuterated Solvents in ^1H NMR**Category: NMR Solvents****COMMON EXAMINER TRAP / MISTAKE:**

Using H_2O or CHCl_3 as an NMR solvent.

MODEL ANSWER / EXAMINER EXPECTATION:

Solvents like H_2O or CHCl_3 contain ^1H protons that would produce huge peaks masking the sample. Deuterated solvents like CDCl_3 or D_2O are used because deuterium (^2H) does not absorb in the ^1H frequency range.

FAQ #9: Singlets for –OH and –NH Protons**Category: ^1H NMR Features****COMMON EXAMINER TRAP / MISTAKE:**

Expecting –OH protons to show splitting from adjacent C–H protons.

MODEL ANSWER / EXAMINER EXPECTATION:

Protons on –OH and –NH groups undergo rapid exchange with trace water/solvents, averaging out spin–spin coupling. Consequently, –OH and –NH protons almost always appear as SINGLETs.

FAQ #10: GC–MS Hyphenated Technique Function**Category: Chromatography–MS****COMMON EXAMINER TRAP / MISTAKE:**

Claiming GC identifies compounds without MS.

MODEL ANSWER / EXAMINER EXPECTATION:

Gas Chromatography (GC) SEPARATES the components of a complex mixture based on retention time. Mass Spectrometry (MS) IDENTIFIES each separated component by its mass spectrum fingerprint.